



CSC 1310-001 Syllabus, Spring 2025

Data Structures and Algorithms

Department of Computer Science, College of Engineering

Instructor Information

April Crockett



Welcome to Data Structures! I am a Senior Lecturer in the Department of Computer Science and have been teaching full-time in the department since 2018. I typically teach the introductory programming course, data structures, and the CS professionalism course that CSC students take during their junior year. I haven't taught 1310 in a few years, and I am excited to teach it again! I am looking forward to an awesome semester with you all.

Email: acrockett@tntech.edu

Office: Bruner Hall Room 330

Office Hours: Mondays and Fridays 11-12pm, or by appointment

You can schedule an appointment by sending me an Email message. I try to respond to messages within 24 hours Monday through Thursday and within 48 hours Friday through Sunday.

Brandon Vandergriff (In charge of CSC 1310 lab sections)



Brandon is the primary instructor for the lab sections in CSC 1310 and teaches one section of lecture. He will be who you contact if there are issues or questions concerning your lab class.

Email: bvandergriff@tntech.edu

Office: Prescott Hall Room 318

Discord: brandonvandergriff

Office Hours: **MW: 9:00am – 9:50am, TR: 9:30am – 12:00pm**

I respond to messages within 24 hours Monday through Thursday and within 48 hours Friday through Sunday.

Eddie Gannod – lead GTA for this section



Hello all! I recently graduated from Tech and am pursuing a Master's Degree. I enjoy video games, books, and playing the saxophone. I am passionate about education and helping people to understand new concepts, and I'm excited to work with everyone!

Email: epgannod42@tntech.edu

Office Hours in TA Tutoring Space: Clement Hall 402

Office Hours: Tuesdays 4:30pm-5:30pm, Wednesdays 2:00pm-3:30pm, Thursdays 4:30pm-6:00pm

Course Logistic Information

Location: Bruner Hall, Room 228

Days: Tuesday/Thursday

Time: 1:30pm – 2:45pm

Lab Class:

You are required to attend both lecture class and your lab class. The course master schedule (in ilearn under Content) lists the weeks that you will attend lab, and the first lab class is week of **Monday, January 27th, 2025**. The labs have been combined in iLearn.

Below is a list of the six lab sections, times, days, and locations.

LAB SECTION	TIME	DAY	LOCATION
CSC 1310-101	1:00pm-2:15pm	M	AIEB 260
CSC 1310-102	1:00pm-2:15pm	W	AIEB 260
CSC 1310-103	1:00pm-2:15pm	F	AIEB 260
CSC 1310-104	9:30am-10:45am	T	AIEB 260
CSC 1310-105	9:30am-10:45am	R	AIEB 260
CSC 1310-106	4:30pm-5:45pm	T	AIEB 260
CSC 1310-107	4:30pm-5:45pm	R	AIEB 260

Course Overview

Course Description & Goal

Abstract data types and fundamental data structures including stacks, queues, and trees; algorithms to search, sort, and manipulate data using such structures; and introduction to runtime analysis. Students complete a series of weekly laboratory exercises for developing proficiency in implementing and utilizing data structures.

Requirements to Take the Course

Course Prerequisites/Corequisites

Students must meet one of the requirements below.

- Minimum Grade of 'C' in CSC 1300 Intro to Programming
- Minimum grade of 'C' in MATH 1910 Calculus I – may be taken concurrently.

Technology Requirements

- Laptop with internet access. Students should follow the College of Engineering recommended Computer Specifications <https://www.tntech.edu/engineering/programs/resources/comp-specs.php>
- Follow the "Getting your computer set up for CSC 1310" document to get the required software for the course, which includes a text editor and C++ compiler.

Time Requirements

- This course will require a time commitment of 4 hours of classroom & lab time, plus approximately 7-10 hours of "out-of-class" work for a total of 11-14 hours per week.

Major Teaching Methods

- Lectures
- Programming assignments
- Lab assignments

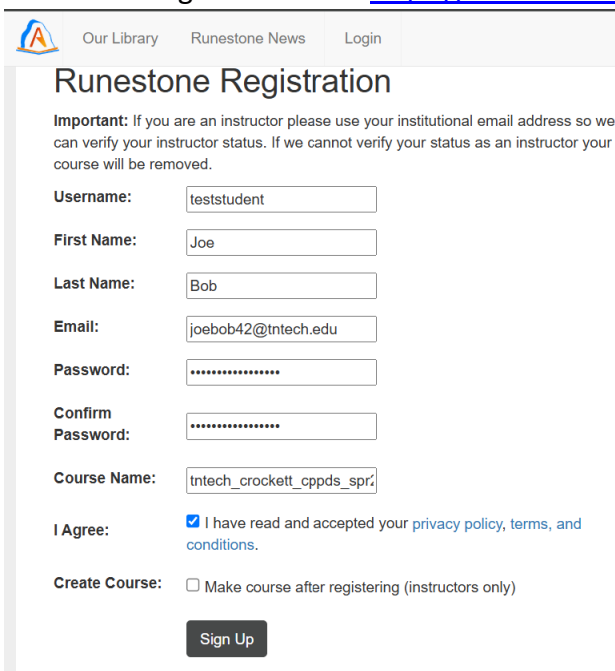
Required Materials & Instructional Platforms

Required Textbook

This section is using an open education resource online textbook in Runestone Academy. The title is **Problem Solving with Algorithms and Data Structures using C++**. You will need to register with Runestone Academy for this textbook, but it is free!

Register with Runestone Academy using the link below.

1. Refer to the provided screen shot below.
2. Note that when registering, your username can be anything, but the **email needs to be your Tennessee Tech email address**.
3. Make sure your **name matches the name that is in iLearn**.
4. Enter this text string as your course name: **tntech_crockett_cppds_spr25**
5. Here is the registration link: <https://runestone.academy/runestone/default/user/register>



The screenshot shows the Runestone Academy registration page. At the top, there is a navigation bar with a logo and links for 'Our Library', 'Runestone News', and 'Login'. The main heading is 'Runestone Registration'. Below this, an important note states: 'Important: If you are an instructor please use your institutional email address so we can verify your instructor status. If we cannot verify your status as an instructor your course will be removed.' The form contains several input fields: 'Username' (filled with 'teststudent'), 'First Name' (filled with 'Joe'), 'Last Name' (filled with 'Bob'), 'Email' (filled with 'joebob42@tntech.edu'), 'Password' (masked with dots), 'Confirm Password' (masked with dots), and 'Course Name' (filled with 'tntech_crockett_cppds_spr25'). There is a checkbox for 'I Agree' which is checked, with a link to 'privacy policy, terms, and conditions'. Below that is a checkbox for 'Create Course' with the label 'Make course after registering (instructors only)'. At the bottom of the form is a 'Sign Up' button.

Optional Resources

- <https://visualgo.net/en> - visualizing data structures and algorithms through animation
- <https://www.geeksforgeeks.org/data-structures/?ref=shm>
- <https://leetcode.com/explore/featured/card/leetcode-interview-crash-course-data-structures-and-algorithms/>

iLearn (TNTECH Learning Management System)

Tennessee Tech's learning management system, iLearn <https://elearn.tntech.edu>, is where all announcements, course materials, and sample programs are for this course.

Turn on notifications in iLearn to ensure you do not miss any due dates.

- Set up your Notifications (video): <https://www.youtube.com/watch?v=JyVYtlzpoI0>
- Set up your Notifications (pdf): <https://www.tntech.edu/ilearn/pdf/ilearn-notifications.pdf>

- Download the Brightspace Pulse App on your phone for quick access to course materials, schedules, and notifications.
 - Apple App Store: <https://apps.apple.com/us/app/brightspace-pulse/id1001688546>
 - Google Play: <https://play.google.com/store/apps/details?id=com.d2l.brightspace.student.android>

Topics to be Covered

- | | | |
|------------------------|-----------------------|------------------------|
| • Objects/Classes | • Parallel Algorithms | • Binary Search Tree |
| • Namespaces | • Lists | • Graphs |
| • Templates | • Stacks | • Balanced Trees / AVL |
| • Recursion | • Queues | • Heaps |
| • Algorithm Analysis | • Hashing | • Priority Queue |
| • Searching Algorithms | • Maps | • Sets |
| • Sorting Algorithms | • Trees | |

Course Schedule

The detailed course schedule is provided in a separate document named “**CSC 1310 Spring 2025 Master Schedule**”.

Course Learning Objectives

1. Students can apply the tools available for problem solving and algorithm development.
2. Students will demonstrate the ability to identify and use the appropriate data structure(s) for a given problem.
3. Students will be able to perform testing and debugging of programs.
4. Students will be able to develop, design, and document intermediate procedural and object-oriented programs in C++.
5. Students will have the ability to perform the runtime analysis of a program.
6. Students will be able to read, critique (improve), and verify the correctness of code.
7. Students will be able to construct precise and effective prompts for large language models (LLMs) to generate code that correctly implements efficient algorithms to solve computational problems.
8. Students will have the ability to implement algorithmic solutions that are representative of data parallelism.
9. Students will demonstrate the ability to identify when a problem solution requires interaction with a remote service.
10. Students will be able to develop algorithmic solutions for interacting with a remote service.
11. Students will have the ability to recognize when an algorithmic solution can benefit from the use of multiple threads that communicate via events or similar means of interaction.

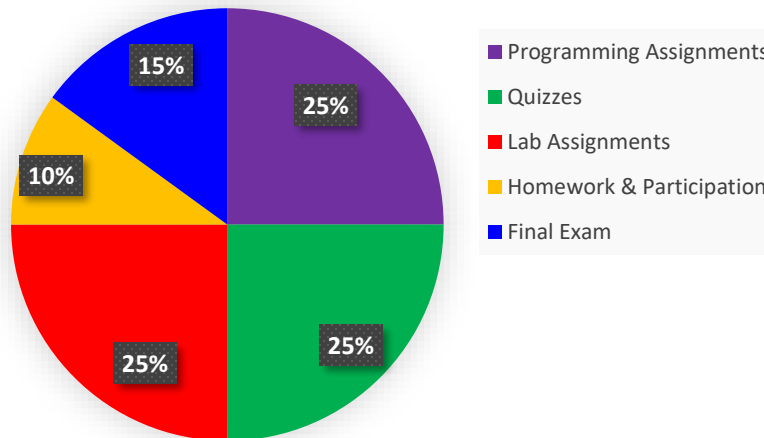
Grading & Evaluation Procedures

Grade Distribution

- | | |
|----|----------------|
| A: | 90.0 to 100 |
| B: | 80.0 to 89.9 |
| C: | 70.0 to 79.9 |
| D: | 60.0 to 69.9 |
| F: | 59.9 and below |

Grade Distribution by Assessment Type

Assessment Type	Percentage of Overall CSC 1300 Grade
Programming Assignments	25%
Quizzes	25%
Lab Assignments	25%
Homework & Participation	10%
Final Exam	15%



- **Programming Assignments** – there will be four major programs assigned in this class
- **Quizzes** – there will be several quizzes during class covering lecture modules. You will have to write code without notes or online help on these quizzes.
- **Final Exam** – there will be a comprehensive final exam
- **Lab Assignments** – weekly assignments given to you in lab – part of this grade will require a report to be submitted.
- **Homework & Participation** – Runestone Academy textbook homework assignments and in-class participation/assignments which may include watching a few videos

Course Policies

Grading Policy

- If you have an issue with any grade posted, you must contact me via email **within 2 days of the grade being released**. If you have notifications set up, you will know all your grades when they are posted.
- All grades will be posted to ilearn.
- Final grades are not rounded up. Example: If you earned an 89.9, then your final grade is a 'B'.
- **All emails asking us to add points to your grade at the end of the semester will not be answered.**
- Programming assignments will be graded within 1.5-2 weeks of completion.
- All lab and programming assignments will have rubrics that show how you will be graded. Always pay attention to the grade rubric prior to the due date.

Attendance Policy

Attendance is required for both lab and lecture. Make sure to read the **late quiz policy** below.

Students who are unable to attend class for an extended period due to an emergency/extenuating circumstance (i.e., medical illness, hospitalization, death in the family/bereavement, military or legal obligation), may contact the Office of the Vice President for Student Affairs at studentaffairs@tntech.edu to request an absence notification.

Late Quiz Policy

- You must have a documented (doctor's note), excused absence to make up a missed quiz.
- You must email Mrs. Crockett within **ONE DAY** of missing a quiz to be eligible to make up the missed quiz.
- You will know which days there is a quiz because quiz dates are in the master schedule.

Late Assignment Policy

- **Lecture programming assignments** may be completed/submitted **up to 2 days late** at a penalty of **10 points off per day late**.
- **After 2 days late, the assignment may not be submitted**, and the grade will be a zero. **Emailed submissions are not accepted.**

Class Participation

In this course, the instructors hope to foster a climate of mutual respect where we learn and communicate together as a community where diverse perspectives are heard. Below is a list of expectations to ensure productive and effective discussions:

- It is expected that every student will **participate actively, maturely, and respectfully** in course discussions.
- Assume that others have the best intentions.
- Recognize and value the diverse backgrounds, experiences, abilities, and knowledge each person brings to the class.

Email Policy

You will be expected to have your notifications set up in ilearn to receive an email when an announcement has been posted to your courses. You will be expected to check your Tennessee Tech email account at least once a day.

Student Academic Misconduct

Maintaining high standards of academic integrity in every class is critical to the reputation of Tennessee Tech, its students, faculty, alumni, and the employers of Tennessee Tech graduates. Academic integrity is at the foundation of the educational process and key to student success. Students with academic integrity are committed to honesty, ethical behavior, and avoiding academic integrity violations. All students must read and understand Policy 216: Student Academic Integrity. Please see the Academic Integrity website (<https://www.tntech.edu/provost/academicintegrity/>) for more information.

The Department of Computer Science considers cheating to be a very serious offense, and this conduct will result in a zero on the assignment as well as a submission for academic misconduct for all involved parties. Here is a list of activities that are considered cheating (Academic Misconduct) in CSC 1300 for individual assignments:

1. Showing another CSC 1310 student more than one to two lines of your code is cheating. This includes emailing (or otherwise sending) your code to someone or looking at code that was emailed (or

otherwise sent) to you. The only exception is if the other student is your peer programming partner in a lab assignment.

2. Having your mom/dad/friend/person help you with any part of your programming assignment who are not one of the TAs or an instructor and not providing a comment saying that you got help from that person on a part of your assignment. Note that getting help from outside sources is not encouraged as they may not be familiar with the concepts and up-to-date versions of C++ that we are teaching.
3. Using generative artificial intelligence (ChatGPT, Copilot, etc) to create or assist in creating your program when it is not allowed. (See the Generative AI Policy below)
4. Taking solutions off any website or forum and submitting them as your own work. You **MUST** document all code sources if you use any part of code from an external source outside of Zybooks, course presentation slides, or example programs provided by the instructors. This is an example of what we should see if you an external resource like stackoverflow:

```
# apparently cannot negate df[df['a']=='something']  
# have to union and remove duplicates  
# https://stackoverflow.com/questions/38681340/how-to-remove-common-rows-in-two-dataframes-in-pandas
```

5. Paying someone to write your code.

You ARE encouraged to search for solutions to your problems online via resources discussed in class (cplusplus.com, stackoverflow, or geeksforgeeks) but do not copy/paste any code and **always reference a source in your program comments if you use it!**

Generative AI Policy

In this course, Generative AI resources are allowed to be used for specific assignments, as designated by the instructor. Each assignment where generative AI is allowed, it will be clearly stated. If generative AI is not allowed, it will be clearly stated in the assignment.

To ensure academic integrity, students must openly disclose any AI-generated material they utilize and provide proper attribution. This includes in-text citations, quotations, and references.

To indicate the use of a Generative AI resource, a student should include the following statement in their assignments: "The author(s) acknowledge the utilization of [Generative AI Tool Name], a language model developed by [Generative AI Tool Provider], in the preparation of this assignment. The [Generative AI Tool Name] was employed in the following manner(s) within this assignment [e.g., brainstorming, grammatical correction, citation, specific section of the assignment]."

Proper citation guidelines can be found on the [CITL website](#).

Policy Concerning What You Can't Use in Programming Assignments

You may not use code or syntax in your programming assignments that have not been taught in the class unless otherwise specified (example programs, presentation slides, assignments).

You may not ever use the following in your programs:

- while(true) or while(1)
- Goto statements
- Recursively calling the main function
- exit()
- ternary operators (any syntax requiring a '?')

If you use any of the things listed above or use code that has not been taught in the class (without referencing your detailed source) then full points will be deducted for that criteria item in the rubric.

Disability Accommodation

Students with a disability requiring accommodations should contact the accessible education center (AEC). An accommodation request (AR) should be completed as soon as possible, preferably by the end of the first week of the course. The AEC is in the **Roaden University Center, room 112**; phone **931-372-6119**. For details, view Tennessee Tech's policy 340 – [services for students with disabilities at policy central](#).

Let Others Get to Know You

One way that you can help build community between students and the instructor in this course is by letting others get to know you. Below are instructions on how you can change your name to your preferred name in iLearn and add your profile photo in iLearn and in your school email.

Change to Your Preferred Name in iLearn!

<https://blogs.tntech.edu/techtimes/2022/administration/how-to-preferred-name-change-for-students/#:~:text=Students%20can%20request%20to%20use,Just%20fill%20out%20the%20form>.

Add Your Profile Photo to iLearn

1. Click on your name in the top-right corner of ilearn once you are logged in.
2. Click **“Profile”**
3. Click **“Change Picture”**

Add Your Profile Photo to Microsoft 365

One helpful thing you can do to let your instructors or potential future employers get to know you is to add a profile picture to your TTU email in outlook. It is easy. Here are directions on how to do it:

<https://support.office.com/en-us/article/add-your-profile-photo-to-office-365-2eaf93fd-b3f1-43b9-9cdc-bdcd548435b7>

Additional Campus Resources

Technical Help

If you are experiencing technical problems, visit the [myTech IT Helpdesk](#) for assistance.

If you are having trouble with one of the instructional technologies (i.e. Zoom, Teams, Qualtrics, iLearn Respondus, or any technology listed [here](#)) visit the [Center for Innovation in Teaching and Learning](#) (CITL) website or call 931-372-3675 for assistance.

For accessibility information and statements for our instructional technologies, visit the [CITL's Learner Success Resource page](#).

Tutoring

The university provides free online and in-person tutoring to all Tennessee Tech students – including in Computer Science classes. Tutoring is available for any class or subject, as well as writing, test prep, study skills, and resume support. Appointments are scheduled, so contact the [Learning Center website](#) for more information.

Health and Wellness

Counseling Center

The Counseling Center offers brief, short-term, solution-focused therapeutic interventions for Tennessee Tech University students. The staff of the Counseling Center is available to assist students with their personal and social concerns in hopes of helping them achieve satisfying educational and life experiences. To learn more or schedule an appointment, visit the [Counseling Center website](#).

Health Services

Health Services offers high-quality, affordable care that is accessible and promotes the health and wellness of our Tennessee Tech community. Visit the [Health Services](#) website to learn more.

Pandemic Protocols

Each student must take personal responsibility for knowing and following any University protocol related to pandemics and other public health events. Students are expected to follow all directives published by Tennessee Tech on its official webpage. As conditions related to the COVID-19 pandemic change, the University's COVID-19 protocols are also likely to change. Students are expected to monitor the University's official webpage to stay up to date on public health protocols.

Syllabus & Schedule are Tentative

The syllabus and schedule are subject to change.